

Evaluating asymptotic intervals of FH estimates with and without variance smoothing, using simulated data mirroring Kenya height-for-age

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2025-12-01

The goal of this simulation study is to evaluate how well variance smoothing can improve interval properties under the FH model. We simulate data meant to mirror the height-for-age z-score in the 2022 Kenya DHS.

Simulation Design

1. Obtain weighted stratified mean estimates $(\boldsymbol{\mu}_R, \boldsymbol{\mu}_U)$, at the admin2 level, using the DHS data (height-for-age)
2. Generate a continuous response, for each sampled individual k in cluster c , in strata h , in admin 2 area j , in admin 1 area i , from either
 - a. no cluster effect:

$$y_{ijhck}^{(s)} \sim \mathcal{N}(\mu_{jh}, \sigma_{jh}^{2(s)})$$

- b. cluster effect:

$$y_{ijhck}^{(s)} \sim \mathcal{N}(\mu_{jh} + \epsilon_c^{(s)}, \sigma_{jh}^{2(s)})$$

where $\sigma_{jh}^{2(s)}$ is generated from,

$$\log(\sigma_{jR}^{2(s)}) = \alpha + Z\gamma + v_j$$

$$\sigma_{jU}^{2(s)} = (1 + \kappa_{i[j]})\sigma_{jR}^{2(s)}$$

where Z is a covariate matrix, and v can be distributed either IID or BYM2.

3. Take a two-stage stratified sample, with sampling design and sizes chosen to mirror the Kenya 2022 DHS.
4. Fit models (admin1 and admin2 level) using a mean model with BYM2 effects and covariates and the following variance models:
 - a. IID variance and constant strata effect:

$$\log(\sigma_{iR}^{2(s)}) \sim \mathcal{N}(\alpha, 1/\tau)$$

$$\sigma_{iU}^{2(s)} = (1 + \kappa)\sigma_{iR}^{2(s)}$$

- b. Structured variance and constant strata effect:

$$\log(\sigma_{iR}^{2(s)}) = \alpha + Z\gamma + v_i$$

$$\sigma_{iU}^{2(s)} = (1 + \kappa)\sigma_{iR}^{2(s)}$$

where Z is a covariate matrix, and v can be distributed either IID or BYM2

- c. IID variance and varied strata effect:

$$\log(\sigma_{iR}^{2(s)}) \sim \mathcal{N}(\alpha, 1/\tau)$$

$$\sigma_{iU}^{2(s)} = (1 + \kappa_i)\sigma_{iR}^{2(s)}$$

where κ_i is always an admin1 level parameter, regardless of model level

- d. Structured variance and varied strata effect:

$$\log(\sigma_{iR}^{2(s)}) = \alpha + Z\gamma + v_i$$

$$\sigma_{iU}^{2(s)} = (1 + \kappa_i)\sigma_{iR}^{2(s)}$$

5. Compare each model to the oracle model (where sampling variance estimate is correct), standard Fay-Herriot (no smoothing) and the naive variance smoothing model with $\text{df} = \# \text{clusters} - \# \text{strata}$. The naive variance smoothing model assumes that cluster means within a domain are distributed IID, so we will use the following variance models:

- a. IID variance:

$$\log(\sigma_i^{2(s)}) \sim \mathcal{N}(\alpha, 1/\tau)$$

- b. Structured variance:

$$\log(\sigma_i^{2(s)}) = \alpha + Z\gamma + v_i$$